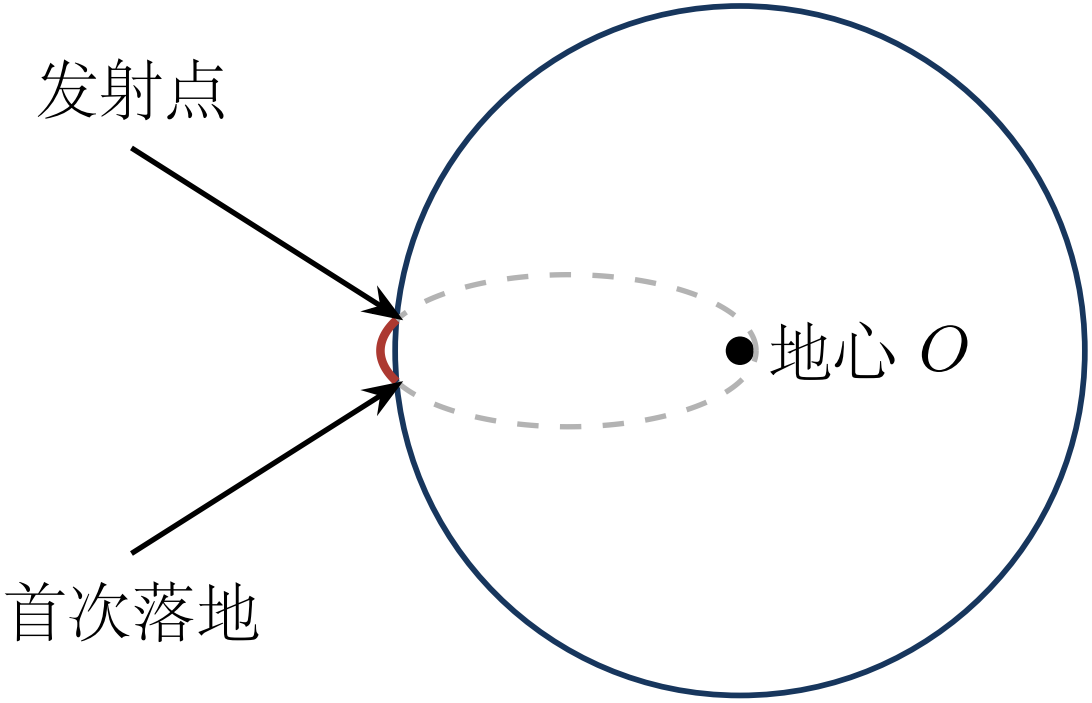
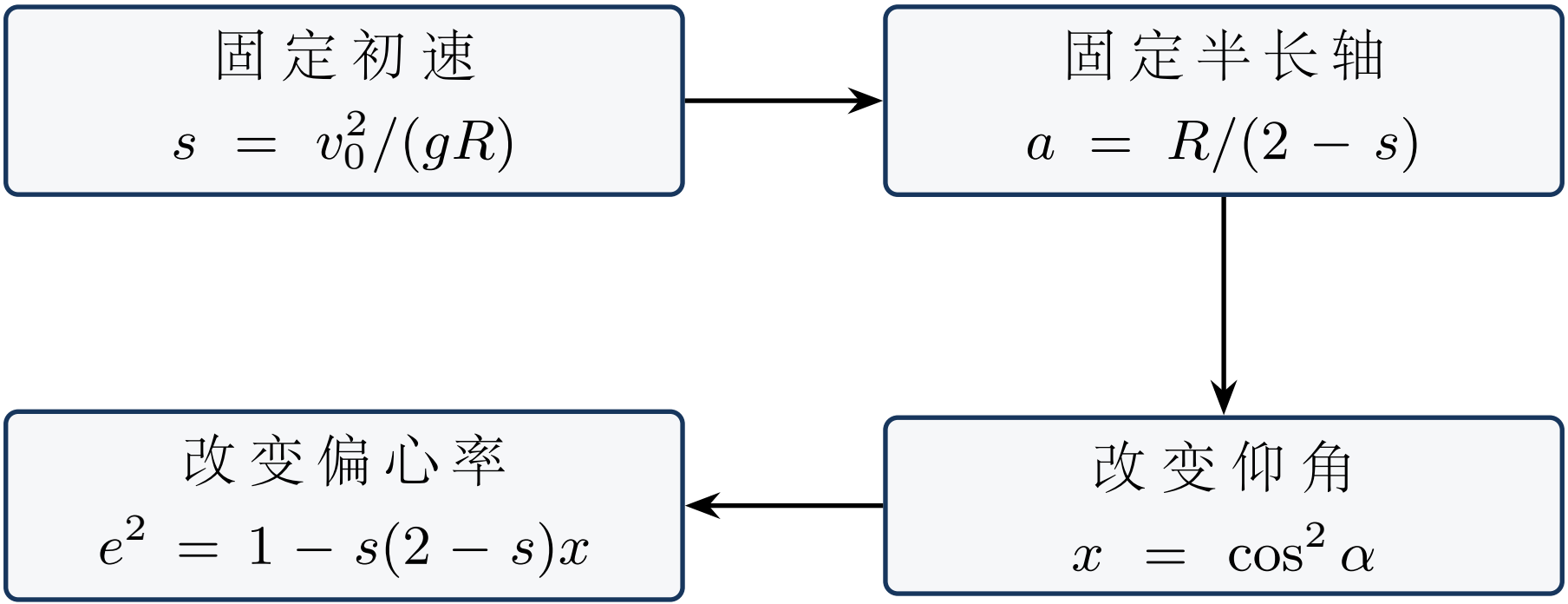
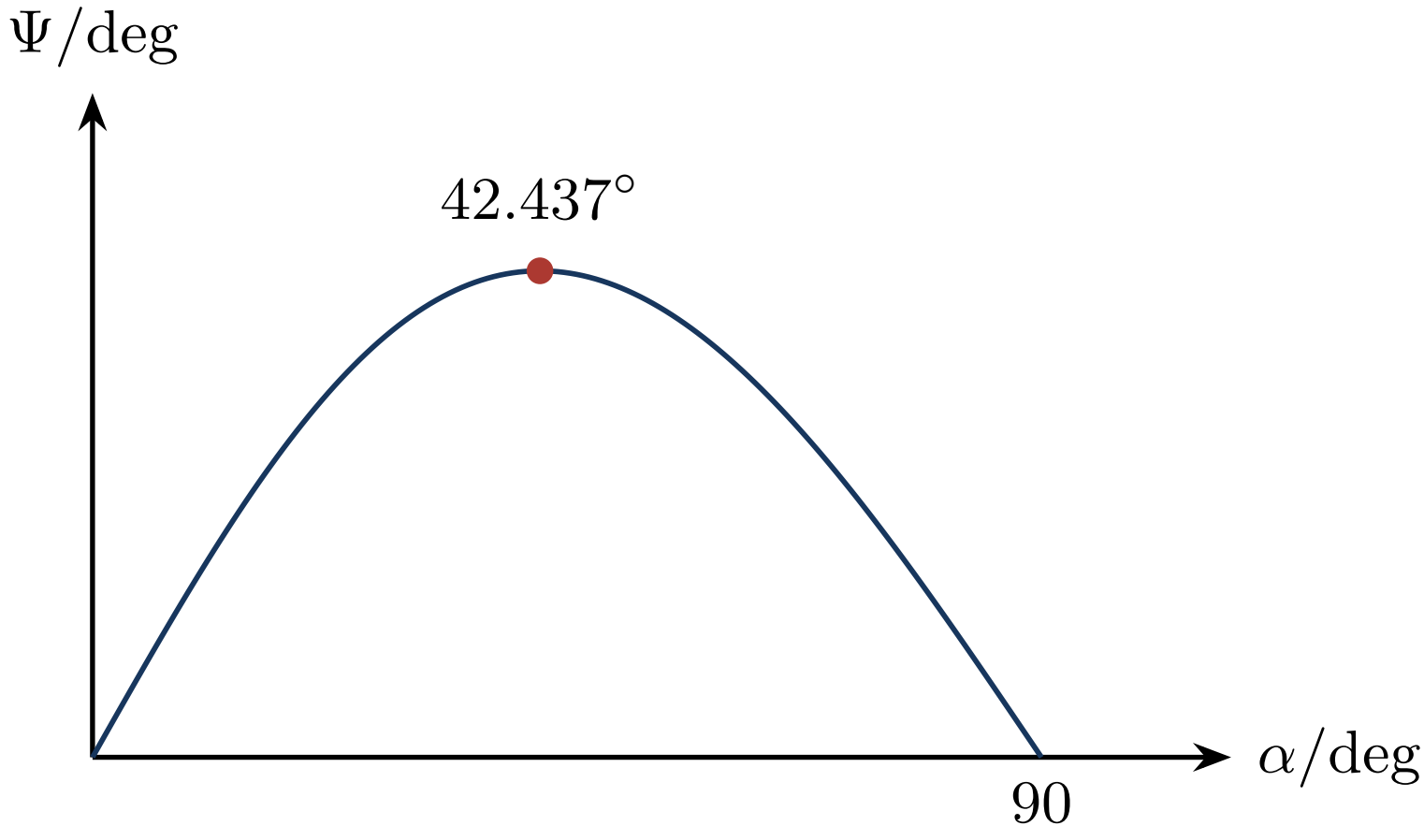


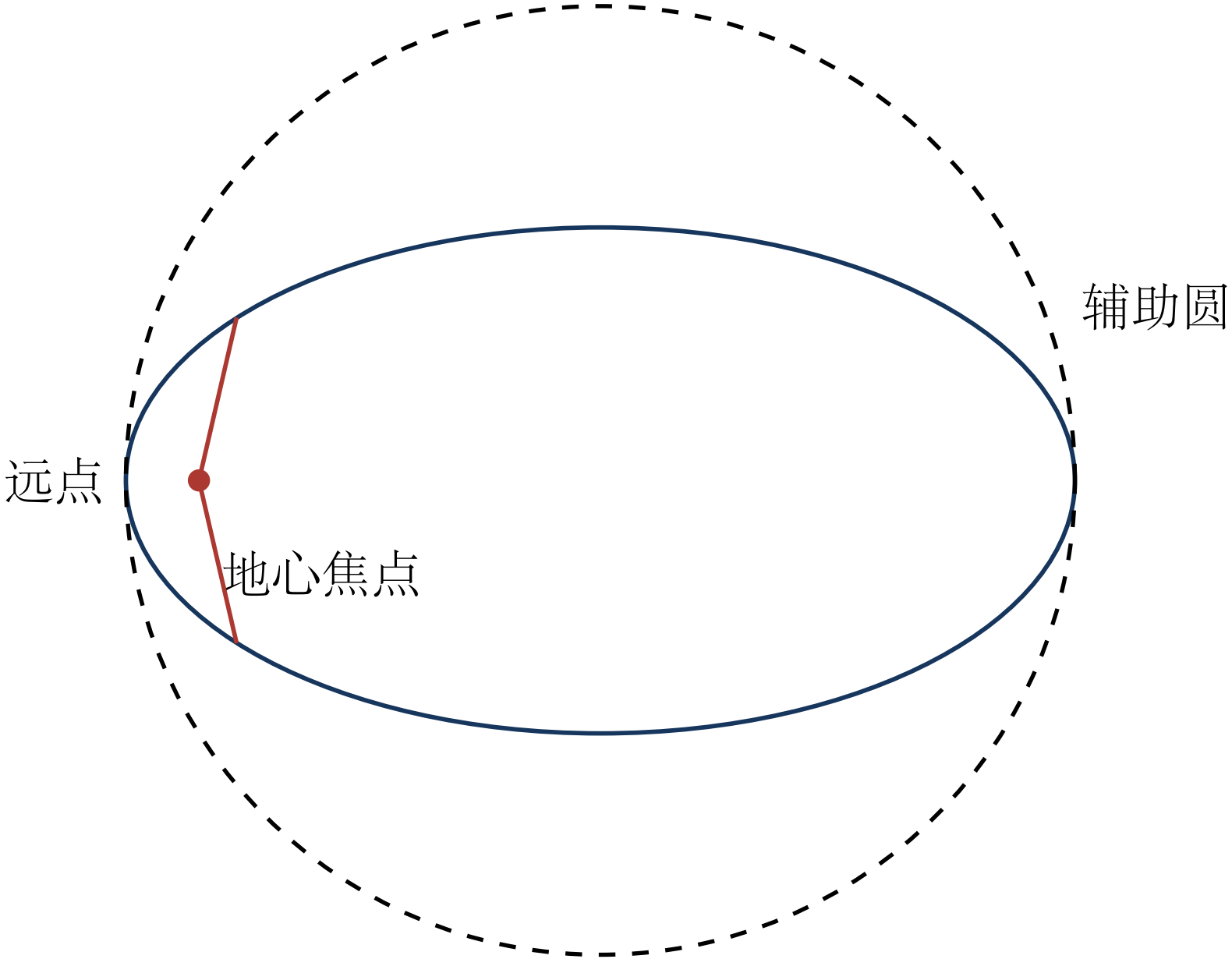
椭圆虚线为数学延长；只取地表以上的红色轨道段



经纬度只定位发射点；球对称且无自转时，方位角任意

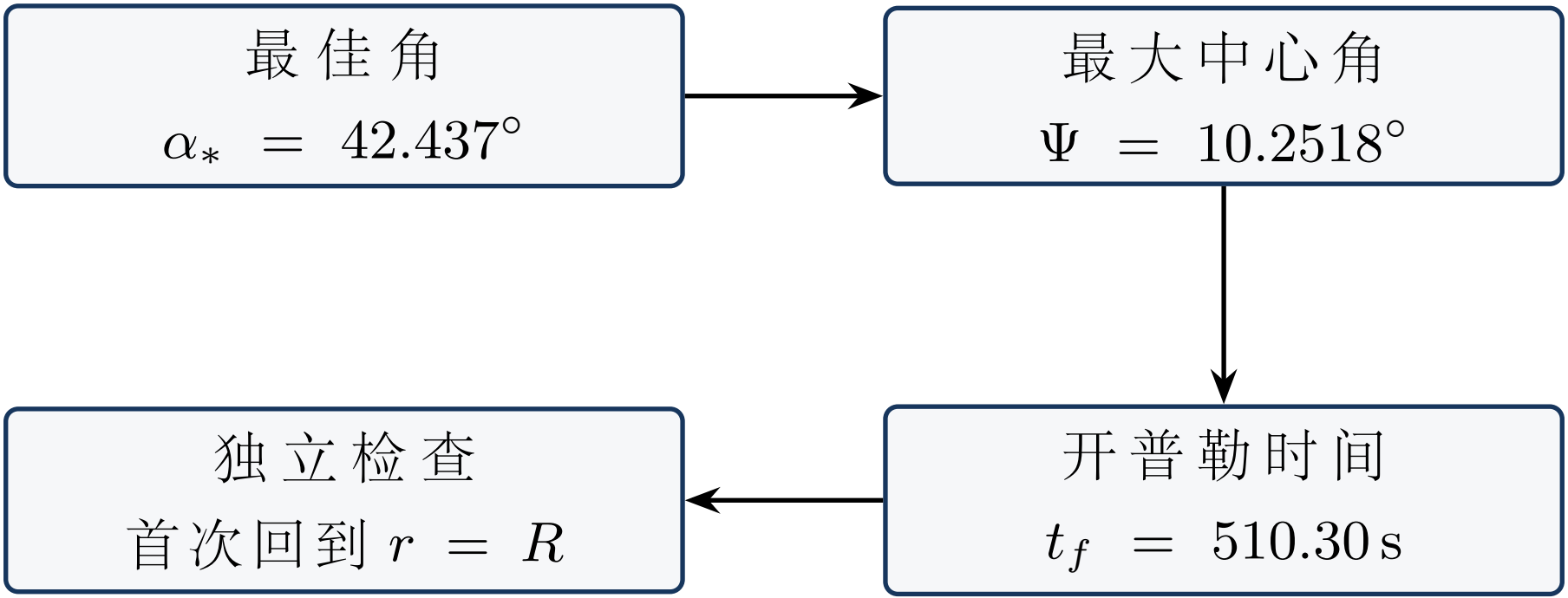


唯一内部最大值；两端中心角均为零



偏近点角区间 $[\pi - \eta, \pi + \eta]$

焦点扫面积含 $\eta + e\sin\eta$；本图为参数示意



平抛极限：$\alpha_* \to 45^\circ$，$t_f \to \sqrt{2}v_0/g$